

General Organic Chemistry (GOC) Comprehensive Notes

Class 11 Foundation & Advanced Electron Effects

Chapter / Focus Area: Basic Principles and Techniques in Organic Chemistry

1. Electronic Displacement Effects

- Inductive Effect (+I / -I): Permanent, sigma-bond polarization. Decreases rapidly with distance.

-I series: $-\text{NO}_2 > -\text{CN} > -\text{COOH} > -\text{F} > -\text{Cl} > -\text{Br} > -\text{I} > -\text{OH} > -\text{NH}_2 > -\text{C}_6\text{H}_5 > -\text{H}$

+I series: $-\text{O}(-) > -\text{COO}(-) > -\text{C}(\text{CH}_3)_3 > -\text{CH}(\text{CH}_3)_2 > -\text{CH}_2\text{CH}_3 > -\text{CH}_3 > -\text{T} > -\text{D} > -\text{H}$

2. Resonance & Mesomeric Effects

- Delocalization of pi-electrons in conjugated systems.
 - +M groups (+R): $-\text{OH}$, $-\text{NH}_2$, $-\text{OCH}_3$, $-\text{Cl}$ (ortho/para-directing in EAS).
 - M groups (-R): $-\text{NO}_2$, $-\text{CHO}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$ (meta-directing, strongly deactivating).
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3. Reaction Intermediates & Stability Order

- Carbocations (sp^2 , planar): Tropylium $>$ Benzyl $>$ Allyl $>$ $3^\circ > 2^\circ > 1^\circ >$ Methyl.
 - Carbanions (sp^3 , pyramidal): Methyl $>$ $1^\circ > 2^\circ > 3^\circ$ (stabilized by -I/-M groups).
 - Free Radicals ($\text{sp}^2/\text{planar}$): $3^\circ > 2^\circ > 1^\circ >$ Methyl (stabilized by hyperconjugation).
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Official Verified Study Resource • Mentored by Ajay Choudhary Sir

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